

Advanced Subnetting & VLSM

1. Given a company with 3 departments (HR: 20 users, Sales: 50 users, Tech: 100 users), assign appropriate subnets using VLSM from the 192.168.10.0/24 network and write the subnet address, subnet mask, and usable IP range for each department.

->Using **VLSM (Variable Length Subnet Masking)**, assign the largest department first and then allocate smaller subnets.

Step 1: Determine subnet sizes

- **Tech: 100 users** → /25 → 126 usable IPs
- **Sales: 50 users** → /26 → 62 usable IPs
- **HR: 20 users** → /27 → 30 usable IPs

VLSM Allocation

Department	Users	CIDR	Subnet Mask	Network Address	Usable IP Range	Broadcast
Tech	100	/25	255.255.255.128	192.168.10.0	192.168.10.1 – 192.168.10.126	192.168.10.127
Sales	50	/26	255.255.255.192	192.168.10.128	192.168.10.129 – 192.168.10.190	192.168.10.191
HR	20	/27	255.255.255.224	192.168.10.192	192.168.10.193 – 192.168.10.222	192.168.10.223

Why these subnet sizes?

- **Tech:** /25 provides 126 usable addresses, enough for 100 users.
- **Sales:** /26 provides 62 usable addresses, enough for 50 users.
- **HR:** /27 provides 30 usable addresses, enough for 20 users.

This uses addresses through **192.168.10.223**, leaving **192.168.10.224–192.168.10.255** available for future expansion.

2. You are given three office locations (Ahmedabad, Surat, Rajkot). Each office has three departments: Admin (10 users), Marketing (30 users), and Support (60 users). Design an IP addressing scheme using VLSM for all offices using the 10.0.0.0/22 block, ensuring no overlap and efficient usage. List each subnet with its office, department, subnet mask, and first/last usable IP.

->Using **VLSM**, we allocate the largest departments first and use progressively smaller subnets for smaller departments.

The available block is **10.0.0.0/22**, which covers:

10.0.0.0 – 10.0.3.255 (1,024 total addresses)

Required subnet sizes

- **Support: 60 users** → /26 → 62 usable addresses
- **Marketing: 30 users** → /27 → 30 usable addresses
- **Admin: 10 users** → /28 → 14 usable addresses

A convenient allocation is to give each office one /25-sized section, containing its three VLSM subnets.

Office	Department	CIDR	Subnet Mask	First Usable IP	Last Usable IP
Ahmedabad	Support	10.0.0.0/26	255.255.255.192	10.0.0.1	10.0.0.62
Ahmedabad	Marketing	10.0.0.64/27	255.255.255.224	10.0.0.65	10.0.0.94
Ahmedabad	Admin	10.0.0.96/28	255.255.255.240	10.0.0.97	10.0.0.110
Surat	Support	10.0.0.128/26	255.255.255.192	10.0.0.129	10.0.0.190
Surat	Marketing	10.0.0.192/27	255.255.255.224	10.0.0.193	10.0.0.222
Surat	Admin	10.0.0.224/28	255.255.255.240	10.0.0.225	10.0.0.238
Rajkot	Support	10.0.1.0/26	255.255.255.192	10.0.1.1	10.0.1.62
Rajkot	Marketing	10.0.1.64/27	255.255.255.224	10.0.1.65	10.0.1.94
Rajkot	Admin	10.0.1.96/28	255.255.255.240	10.0.1.97	10.0.1.110

Summary

- **Support:** /26 → 62 usable IPs
- **Marketing:** /27 → 30 usable IPs
- **Admin:** /28 → 14 usable IPs
- **Total used:** 336 addresses across all three offices.
- **No subnet overlaps.**
- The remaining portion of **10.0.0.0/22** is available for future departments, servers, expansion, or additional offices.

This scheme meets every department's minimum requirement while avoiding unnecessarily large subnets for the smaller teams.

3. A startup has offices in Mumbai, Pune, and Baroda. Each office will connect to HQ via VPN. Propose a route summarization for all three offices if their assigned subnets are: Mumbai (172.16.10.0/24), Pune (172.16.11.0/24), Baroda (172.16.12.0/24). Write the summarized route and explain why it is efficient.

Hint: Find the smallest supernet that includes all three subnets.

->The three office networks are:

- **Mumbai:** 172.16.10.0/24
- **Pune:** 172.16.11.0/24
- **Baroda:** 172.16.12.0/24

Step 1: Find the smallest supernet

The third octet values are **10, 11, and 12**.

To summarize them with one CIDR block, the smallest convenient supernet is:

172.16.8.0/21

A /21 covers:

172.16.8.0 – 172.16.15.255

Therefore, it includes all three office networks:

- 172.16.10.0/24 
- 172.16.11.0/24 

- 172.16.12.0/24 

Summarized Route

 172.16.8.0/21

Subnet mask: 255.255.248.0

Why is this efficient?

Instead of maintaining **three separate routing entries**, HQ can use **one summarized route**:

172.16.8.0/21

This reduces the size of the routing table and simplifies route management. However, the /21 also includes 172.16.8.0/24, 172.16.9.0/24, 172.16.13.0/24, 172.16.14.0/24, and 172.16.15.0/24, so those ranges should not be assigned to unrelated networks if this summary is advertised.

4. Given a scenario where the Tech department in Ahmedabad grows from 100 to 200 users, update your previous VLSM plan to accommodate the new size with minimal changes to the rest of the network. List the new subnet details and explain your approach.

Constraint: Do not change the existing subnets for HR or Sales unless absolutely necessary.

->Since **Ahmedabad Tech** has grown from **100 to 200 users**, the previous /25 subnet is no longer sufficient.

1. New subnet for Ahmedabad Tech

For 200 users:

- /25 → 126 usable IPs 
- /24 → 254 usable IPs 

So Ahmedabad Tech needs a /24 subnet.

To keep the existing **Ahmedabad Sales and HR subnets unchanged**, we can move only the Tech department to the next available /24 block:

Ahmedabad Tech → 10.0.1.0/24

- Subnet mask: **255.255.255.0**

- Network address: **10.0.1.0**
- First usable: **10.0.1.1**
- Last usable: **10.0.1.254**
- Broadcast: **10.0.1.255**
- Usable addresses: **254**

Updated Ahmedabad plan

Department	Users	CIDR	Subnet Mask	First Usable	Last Usable	Broadcast
Tech	200	10.0.1.0/24	255.255.255.0	10.0.1.1	10.0.1.254	10.0.1.255
Sales	50	10.0.0.128/26	255.255.255.192	10.0.0.129	10.0.0.190	10.0.0.191
HR	20	10.0.0.192/27	255.255.255.224	10.0.0.193	10.0.0.222	10.0.0.223

Approach

The key point is that a /24 cannot fit inside the old Tech 10.0.0.0/25 area without overlapping the existing Sales and HR subnets. Therefore, **only Ahmedabad Tech is moved** to 10.0.1.0/24.

This satisfies the constraint because **HR and Sales remain exactly unchanged**. It also provides Ahmedabad Tech with **254 usable addresses**, leaving room for future growth.

5. Create 5 VLSM subnetting challenge questions inspired by real apps you use (e.g., Zomato delivery zones, Flipkart warehouse locations, or IPL team fan clubs). For each, provide the required number of hosts and a base network address.

->Here are **5 VLSM subnetting challenge questions** based on realistic app/business scenarios. I've included only the **host requirement and base network** so you can solve the subnetting yourself.

1. 🍔 Zomato Delivery Zones

Zomato wants to divide its delivery network into different zones. The largest zone needs **120 hosts**, followed by zones needing **60, 30, and 10 hosts**.

Base network: 192.168.10.0/24

Challenge: Design VLSM subnets for all four zones. Find the CIDR, subnet mask, network address, broadcast address, and usable IP range for each.

2. 📦 Flipkart Warehouse Network

A Flipkart warehouse has three sections:

- Packing: **100 hosts**
- Inventory: **50 hosts**
- Administration: **20 hosts**

Base network: 10.10.0.0/23

Challenge: Use VLSM to allocate the smallest suitable subnet for each section without overlapping addresses.

3. 🖋️ IPL Fan Club Network

An IPL fan-club organization wants separate networks for supporters in three cities:

- Ahmedabad: **200 hosts**
- Mumbai: **100 hosts**
- Surat: **50 hosts**

Base network: 172.20.0.0/22

Challenge: Create an efficient VLSM addressing plan and list the network, broadcast, and usable IP ranges.

4. 🚗 Uber Driver Support Network

A regional Uber office needs networks for:

- Driver Support: **80 hosts**
- Operations: **40 hosts**
- Customer Service: **20 hosts**
- Management: **10 hosts**

Base network: 192.168.50.0/24

Challenge: Apply VLSM and determine the appropriate CIDR prefix and IP range for every department.

5. 🛒 Amazon Regional Office

An Amazon regional office needs four networks:

- Warehouse Systems: **150 hosts**
- Delivery Team: **70 hosts**
- Customer Service: **30 hosts**
- Admin: **15 hosts**

Base network: 10.50.0.0/22

Challenge: Allocate VLSM subnets starting with the largest requirement. Ensure that **none of the subnets overlap** and that there is room for future expansion.